

Prioritizing Business "Levers" Utilizing Linear Regression and Clustering Analysis

Katelyn Patricio

Nevada State University

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Abstract

This capstone explores actionable retail levers to maximize customer value in online grocery retail. Using an online dataset of Instacart customer data with 3.4 million rows, K-Means clustering and regression were applied to uncover four distinct clusters: Inactive Customers, Frequent Shoppers, High Value Customers, and Loyal Low Spenders. This research implemented cluster-specific linear regressions. As a result, customer spend appears to depend heavily on the individual cluster. High Value Customers are highly influenced by basket size, whereas Loyal Low Spenders require aggressive reorder targeting. These insights are provided in a Tableau dashboard (Figure A1) for internal stakeholders, allowing users to filter through various metrics, clusters, and "what-if" scenarios to uncover data-driven recommendations for personalized marketing and customer retention strategies.

Introduction

The retail industry has played a vital role in shaping the global economic landscape, but it has recently seen immense growth in online ecommerce due to a transformation driven by innovative technologies, evolving customer preferences, and economic impacts such as COVID. As digital catalogs continue to improve, platforms such as Instacart face the challenge of moving beyond historical methods (such as mass marketing) to focus on maximizing customer engagement and business profitability.

Modern analytics has shifted towards leveraging customer behavior frameworks to power business models. Identifying patterns in how, when, and why consumers shop provides useful insights into how businesses can improve customer experiences. A paper by Hanyu Wu titled Leveraging Data Analytics and Consumer Insights for Targeted Marketing Campaigns and Personalized Customer Experiences claims that providing customers with "personalized customer experiences often result in higher customer spending" and "can lead to increased average order value and a higher customer lifetime value, contributing to revenue growth" (Wu, 2023). Insights into customer behavior are essential for creating targeted strategies to improve both customer experience and business performance.

Customer lifetime value (CLV) is a key concept in this context. CLV is essentially the total amount of money a customer spends from the first purchase to the last purchase they make with the company, emphasizing the long-term contributions customers make. Achieving and maintaining loyal customers requires identifying actionable "levers" to influence such behavior. In retail analytics, Buy Til' You Die (BTYD) models are employed to forecast customer lifetime

value and probability of re-engagement. However, applying this to this dataset would provide inaccurate data. This is due to a 30-day reporting cap on the 'days_since_prior_order' variable. To overcome this hurdle, a segment-specific regression approach was utilized to provide a comparable perspective on customer value.

This research aims to answer the business question "Which actionable retail levers such as cross-sell opportunities, reorder targeting, and timing strategies should be prioritized to maximize customer value in online grocery retail?" Through the utilization of K-means clustering to identify distinct customer segments and applying linear regressions to each, this project provides a comprehensive framework for identifying key behavioral features informing targeted marketing and customer retention strategies.

Data

The Instacart Market Basket Analysis dataset from Kaggle contains 3.4 million grocery orders from more than 206,000 customers. These files incorporate transactional data including orders, products, and customer purchase history. Descriptive statistics from the data reveal useful insights. Timing information is provided through the 'days_between_orders_average' variable, which shows how often consumers shop, as well as shopping intent through the 'add_to_cart_order' variable. Reorder rate is approximately 59%, displaying strong repeat purchases. There is also an evident hierarchical structure (product → aisle → department), enabling multi-level aggregation, as well as substantial information ensuring reliable coefficients. Despite these strengths, this dataset also has its limitations. No pricing data was included, so a proxy was utilized. It also lacks demographic data, which restricts segmentation of specific behavioral patterns. Furthermore, the 'days_between_order_average' has a 30-day cap, which prevents long-term analysis of customer behavior.

Methodology

This study utilizes a two-step process combining customer segmentation and regression analysis to identify actionable drivers of customer value. In this project, data was imported into R for consolidation and data exploration purposes. An RFM (Recency, Frequency, and Monetary) framework was created by finding distinct orders, number of days between orders, and the average items per order. A consolidated file (the four files joined into one dataset) and an RFM file were exported and brought into Python to apply the customer segmentation and regression analysis. Customers were segmented using K-means clustering based on the RFM framework (Figure A2). In short, K-means clustering leverages an unsupervised machine learning algorithm to segment k distinct groups by similarity. Prior to clustering, an optimal value for K was calculated using the elbow method (Figure A3). Following segmentation, customer-level features were created to capture key behavioral patterns. This included the number of orders placed, average basket size, and average days between orders. These features were selected because they are easily interpretable and are directly related to the purpose of this project. Ridge regression models were then applied to each cluster to determine the relationship between each feature and the target variable 'reorder rate' (Figure A4). A ridge model was employed for its strengths in handling multicollinearity and producing stable coefficients. To further support this, a bootstrap resampling approach was used, allowing standard errors and confidence intervals to be calculated.

Findings

Through segmentation, four distinct clusters were identified: Inactive Customers, Frequent Shoppers, High Value Customers, and Loyal Low Spenders. Linear regressions across each cluster revealed consistent patterns of average basket size and number of orders as the most important drivers of customer value. This indicates a need to increase purchase frequency and order amount in order to improve business performance. Focusing on individual clusters, High Recency customers exhibit focused purchasing behaviors and are primarily influenced by basket size, which suggests an opportunity for Instacart to create personalized, targeted recommendations to increase spending. Low Spend consumers show occasional shopping but seem to lack diversity in category choices. This could be addressed through cross-selling across more aisles. Furthermore, High Value customers display high monetary value and frequent shopping behaviors. It is important to focus on retention and loyalty with this segment. It should be noted that despite all features providing a statistical relationship with customer value, magnitudes differed between each. Those with the strongest relationships should be utilized when making business decisions.

Conclusion

This study demonstrates that combining customer segmentation and cluster-specific regressions provides an effective strategy for identifying key variables of customer value. The findings translate directly to actionable insights and opportunities in online retail. Through this process, I have learned the value that customer data provides. Being able to interpret analytical results into actionable business strategy is a skill I have gained in completing this project. Despite this, if I were to repeat this project, I would incorporate pricing data to provide accurate measurements of customer value. Exploring other models and approaches would also offer deeper insights. In conclusion, through the translation of customer behavior data into actionable insights, this project provides a basis for bridging the gap between analysis and decision-making. Understanding how customers behave is essential for improving customer engagement, increasing retention, and maximizing overall customer value in e-commerce environments.

References

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Appendix Supporting Material

Figure A1. Tableau dashboard

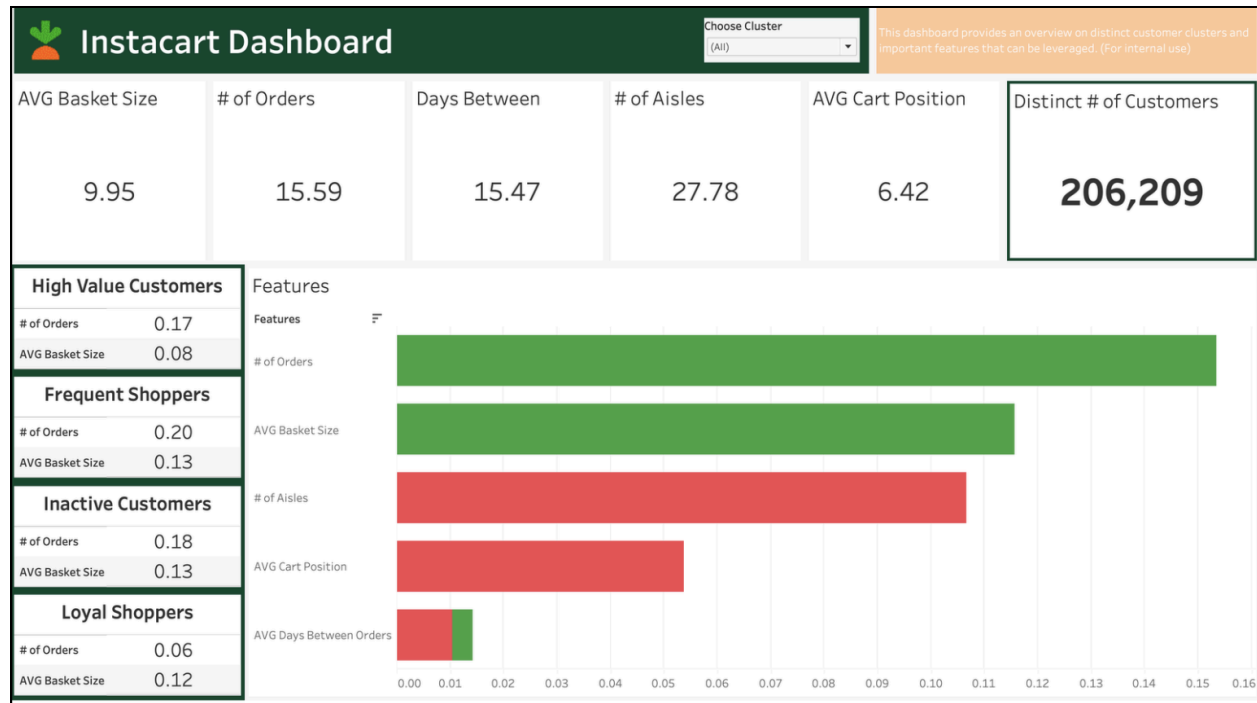


Figure A2. RFM model output aggregated by clusters

	recency		frequency		monetary		count
	mean	median	mean	median	mean	median	
cluster							
0	16.39	15.0	12.03	9.0	19.69	18.40	33593
1	8.32	8.0	12.47	10.0	7.82	7.75	80526
2	27.80	30.0	8.68	6.0	7.75	7.57	70290
3	6.50	6.0	54.88	49.0	9.89	9.36	21800

Figure A3. Results from utilizing elbow method

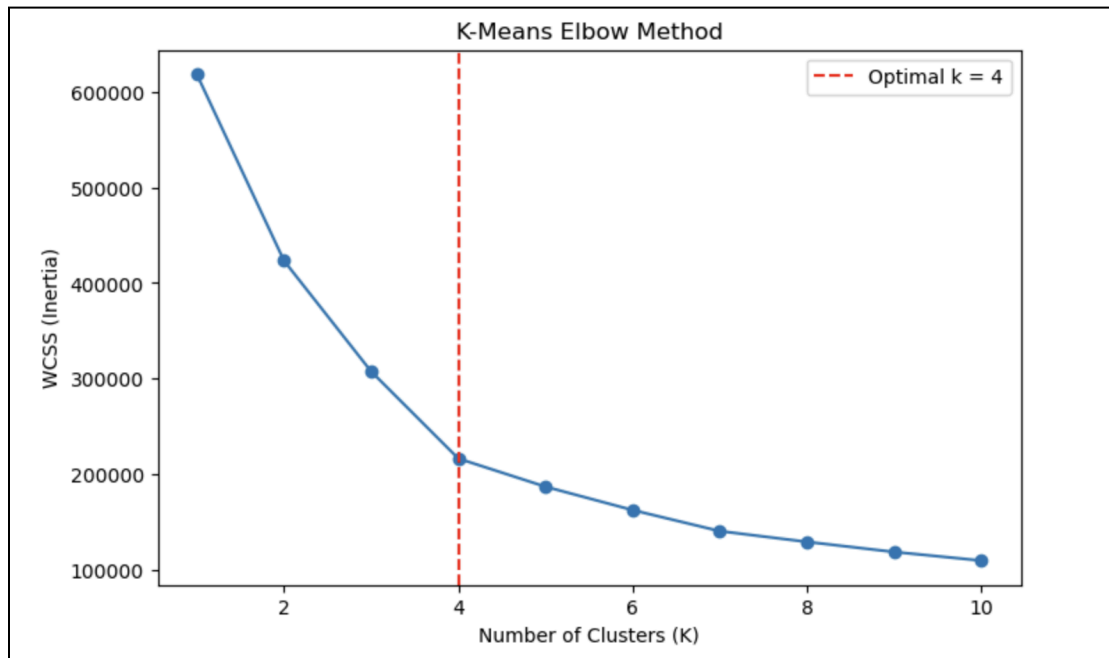


Figure A4. Regression analysis results by cluster

cluster	feature	coef	std_err	lower	upper
0	num_orders	0.170646	0.001016	0.168656	0.172514
0	avg_basket_size	0.077659	0.002047	0.073863	0.081772
0	avg_days_between_orders	-0.011697	0.000840	-0.013310	-0.010124
0	avg_cart_position	-0.049524	0.002122	-0.053726	-0.045492
0	num_aisles	-0.054227	0.001267	-0.056790	-0.051721
1	num_orders	0.201788	0.000618	0.200632	0.203015
1	avg_basket_size	0.130438	0.001456	0.127635	0.133187
1	avg_days_between_orders	0.002951	0.000550	0.001854	0.004015
1	avg_cart_position	-0.058178	0.001435	-0.061120	-0.055384
1	num_aisles	-0.136766	0.000845	-0.138430	-0.135152
2	num_orders	0.177512	0.000934	0.175791	0.179370
2	avg_basket_size	0.130933	0.001683	0.127683	0.134087
2	avg_days_between_orders	-0.008968	0.000761	-0.010475	-0.007558
2	avg_cart_position	-0.063877	0.001602	-0.066949	-0.060706
2	num_aisles	-0.130154	0.001235	-0.132543	-0.127658
3	avg_basket_size	0.124020	0.002356	0.119346	0.128538
3	num_orders	0.063822	0.000963	0.062041	0.065622
3	avg_days_between_orders	0.004703	0.000941	0.002880	0.006563
3	avg_cart_position	-0.043462	0.002478	-0.048441	-0.038626
3	num_aisles	-0.105215	0.000961	-0.107119	-0.103265